SUMMARY – SUPERSTORE DATASET

Source : www.kaggle.com/datasets/vivek468/superstore-dataset-final

Periode : 2014 – 2017

Tujuan : Menganalisa data penjualan superstore untuk mengindentifikasi penjualan, profit, margin dan perkembangan margin dari tahun ke tahun Year of Year (YoY). Sekilas tentang data ini memiliki berapa data transaksi penjualan dari periode tahun 2014 – 2017, dengan total data 9,994 baris, dan 21 kolom.

A. EKSPLORASI & PROFILING DATA

1. Row Count

Total Rows : 9.994
Kolom : 21P
Total Order : 5.069
Total ID Customer : 793
Total Customer : 793
Total ID Product : 1.862
Total Product Name: 1.850
Total City : 531
Total State : 49
Awal Order : 2014 -01-03
Akhir Order : 2017-12-30

Catatan:

Terdapat selisih antara Product ID dan Product Name yaitu 12. Apabila ingin mengetahui Product ID dan Product Name mana saja yang duplikat atau bermasalah perlu dilakukan cek mendalam pada ke dua kolom ini.

B. ANALISA YoY & MODELING

Hasil dari Analisa YoY Tahun 2014-2017 terdapat kenaikan Margin di tahun 2016 namun pada tahun selanjutnya adanya penurunan profit dari 13.43 % ke 12.74%, turun 0,69%. Meskipun penurunan margin terlihat kecil dan masih ke dalam status sehat, penurunan ini mesti harus di evaluasi dan di tinjauan faktor apa yang meyebabkan margin ini turun dan kenapa marginnya bisa turun. Penurunan margin ini dapat di lihat jelas berdasarkan grafik visualisasi di bawah ini:

Period : 2014 – 2017

Objective : Analyse superstore sales data to identify sales, profit, margins, and year-over-year (YoY) margin trends. This dataset contains sales transaction data from the 2014–2017 period, with a total of 9,994 rows and 21 columns.

DATA EXPLORATION & PROFILING DATA

Row Count

Total Rows : 9.994
Column : 21
Total Order : 5.069
Total ID Customer : 793
Total Customer : 793
Total ID Product : 1.862
Total Product Name: 1.850
Total City : 531
Total State : 49
Order Earliest : 2014 -01-03
Order Latest : 2017-12-30

Note:

There is a discrepancy of 12 between the Product ID and the Product Name. To identify which Product IDs and Product Names are duplicates or problematic, need to carry out a thorough check of these two columns.

YOY ANALYSIS & MODELLING

The results of the year-on-year analysis for 2014–2017 show an increase in the margin in 2016; however, in the following year, there was a decline in profit from 13.43% to 12.74%, a decrease of 0.69%. Although the decline in the margin appears small and remains within a healthy range, this decline must be evaluated and the factors causing the margin to fall, as well as the reasons for the decline, must be reviewed. This decline in the margin can be clearly seen in the visualisation graph below:

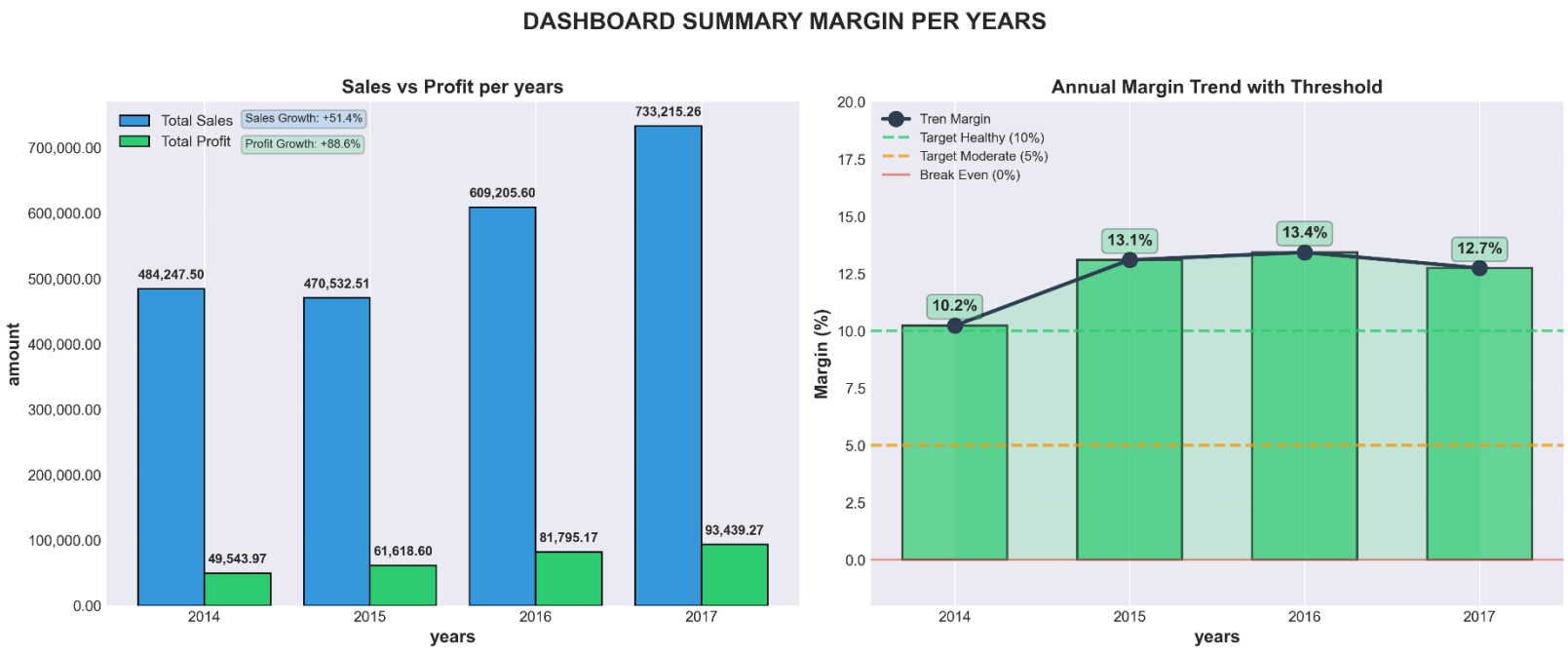
SUMMARY MARGIN PER YEARS

Total Sales : 2,297,200.87
Total Profit : 286,397.01
Total Average: 12.47%

TAHUN	TOTAL SALES	TOTAL PROFIT	MARGIN	STATUS	TOTAL MONTH	AVERAGE MONTH
2014	484,247.50	49,543.97	10.23%	Healthy	12	11.50%
2015	470,532.51	61,618.60	13.10%	Healthy	12	12.21%
2016	609,205.60	81,795.17	13.43%	Healthy	12	13.33%
2017	733,215.26	93,439.27	12.74%	Healthy	12	12.84%

The Best Years : 2016 (13.43%)
The Worst Years: 2014 (10.23%)

Visualisasi data 1 : Sales vs Profit & Annual Margin Trend



Visualisasi data 2 : Growth Sales, Profit & Margin